第9章 结构体与共用体 —— 课后作业答案

基础练习

1. 选择题

题号	答案	解析
(1)	**B** `1`	`s[0].y = 3`，`s[1].x = 2`，`3 / 2 = 1`（整数除法）
(2)	**B** `->`	结构体变量用 `.` 访问成员，结构体**指针**用 `->` 箭头运算符

2. 判断题

题号	答案	解析
(1)	×	共用体所有成员共享同一块内存，同一时间只能存储其中一个成员的值
(2)	?	枚举默认从 0 开始，依次递增。`enum Color {RED, GREEN, BLUE}` 中 RED=0, GREEN=1, BLUE=2

编程练习

3. 日期结构体

```
#include <stdio.h>

typedef struct
{
    int year;
    int month;
    int day;
} Date;

int isLeapYear(int y)
{
    return (y % 4 == 0 && y % 100 != 0) || (y % 400 == 0);
}

int dayOfYear(Date d)
{
    int daysInMonth[] = {0, 31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};
    if (isLeapYear(d.year))
        daysInMonth[2] = 29;
```

```
    int total = 0;
    for (int i = 1; i < d.month; i++)
        total += daysInMonth[i];
    total += d.day;

    return total;
}

int main()
{
    Date d;
    printf("请输入日期(年 月 日): ");
    scanf("%d %d %d", &d.year, &d.month, &d.day);

    printf("%d年%d月%d日是该年的第 %d 天\n",
           d.year, d.month, d.day, dayOfYear(d));
    return 0;
}
```

运行示例：输入 `2024 3 1` → 输出 `第 61 天`（1月31天 + 2月29天 + 1天）

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4. 复数运算

```
#include <stdio.h>

typedef struct
{
    double real;   // 实部
    double imag;   // 虚部
} Complex;

Complex add(Complex a, Complex b)
{
    Complex r;
    r.real = a.real + b.real;
    r.imag = a.imag + b.imag;
    return r;
}

Complex sub(Complex a, Complex b)
{
    Complex r;
    r.real = a.real - b.real;
    r.imag = a.imag - b.imag;
    return r;
}

Complex mul(Complex a, Complex b)
{
    // (a+bi)(c+di) = (ac-bd) + (ad+bc)i
    Complex r;
```

```
    r.real = a.real * b.real - a.imag * b.imag;
    r.imag = a.real * b.imag + a.imag * b.real;
    return r;
}

void printComplex(Complex c)
{
    if (c.imag >= 0)
        printf("%.1f + %.1fi\n", c.real, c.imag);
    else
        printf("%.1f - %.1fi\n", c.real, -c.imag);
}

int main()
{
    Complex a = {3, 2};  // 3+2i
    Complex b = {1, 4};  // 1+4i
    Complex r;

    printf("a = "); printComplex(a);
    printf("b = "); printComplex(b);

    r = add(a, b); printf("a+b = "); printComplex(r);
    r = sub(a, b); printf("a-b = "); printComplex(r);
    r = mul(a, b); printf("a*b = "); printComplex(r);

    return 0;
}
```

运行结果：

```
a = 3.0 + 2.0i
b = 1.0 + 4.0i
a+b = 4.0 + 6.0i
a-b = 2.0 - 2.0i
a*b = -5.0 + 14.0i
```

---

5. 通讯录管理

```
#include <stdio.h>
#include <string.h>
#define MAX 100

typedef struct
{
    char name[20];
    char phone[20];
    char email[30];
} Contact;

Contact contacts[MAX];
```

```c
int count = 0;

void addContact()
{
    if (count >= MAX)
    {
        printf("通讯录已满!\n");
        return;
    }
    printf("姓名: "); scanf("%s", contacts[count].name);
    printf("电话: "); scanf("%s", contacts[count].phone);
    printf("邮箱: "); scanf("%s", contacts[count].email);
    count++;
    printf("添加成功!\n");
}

void searchContact()
{
    char name[20];
    printf("请输入要查找的姓名: ");
    scanf("%s", name);

    for (int i = 0; i < count; i++)
    {
        if (strcmp(contacts[i].name, name) == 0)
        {
            printf("姓名: %s\n", contacts[i].name);
            printf("电话: %s\n", contacts[i].phone);
            printf("邮箱: %s\n", contacts[i].email);
            return;
        }
    }
    printf("未找到该联系人!\n");
}

void listAll()
{
    printf("\n===== 全部联系人 (%d人) =====\n", count);
    printf("%-15s %-15s %s\n", "姓名", "电话", "邮箱");
    for (int i = 0; i < count; i++)
        printf("%-15s %-15s %s\n",
               contacts[i].name, contacts[i].phone, contacts[i].email);
}

int main()
{
    int choice;
    do
    {
        printf("\n1.添加 2.查找 3.显示全部 4.退出\n请选择: ");
        scanf("%d", &choice);
        switch (choice)
        {
```

```
            case 1: addContact(); break;
            case 2: searchContact(); break;
            case 3: listAll(); break;
        }
    } while (choice != 4);
    return 0;
}
```

6. 矩形与圆（共用体）

```
#include <stdio.h>
#define PI 3.14159

typedef struct
{
    int type;  // 0=矩形, 1=圆
    union
    {
        struct { double length, width; } rect;
        double radius;
    } shape;
} Figure;

double area(Figure f)
{
    if (f.type == 0)
        return f.shape.rect.length * f.shape.rect.width;
    else
        return PI * f.shape.radius * f.shape.radius;
}

int main()
{
    Figure f1 = {0, .shape.rect = {5, 3}};  // 矩形
    Figure f2 = {1, .shape.radius = 4};      // 圆

    printf("矩形(5×3)面积 = %.2f\n", area(f1));
    printf("圆(半径4)面积 = %.2f\n", area(f2));
    return 0;
}
```

7. 学生成绩排名（综合）

```
#include <stdio.h>
#define MAX 50

typedef struct
{
    int id;
```

```
    char name[20];
    float score[3];  // 三门课
    float total;
    float average;
    int rank;
} Student;

int main()
{
    Student stu[MAX];
    int n;

    printf("请输入学生人数: ");
    scanf("%d", &n);

    // 输入
    for (int i = 0; i < n; i++)
    {
        printf("\n第%d个学生:\n", i + 1);
        printf("学号: "); scanf("%d", &stu[i].id);
        printf("姓名: "); scanf("%s", stu[i].name);
        printf("三门课成绩: ");
        scanf("%f %f %f", &stu[i].score[0], &stu[i].score[1], &stu[i].score[2]);

        stu[i].total = stu[i].score[0] + stu[i].score[1] + stu[i].score[2];
        stu[i].average = stu[i].total / 3.0;
    }

    // 按总分排序（冒泡，降序）
    for (int i = 0; i < n - 1; i++)
        for (int j = 0; j < n - 1 - i; j++)
            if (stu[j].total < stu[j + 1].total)
            {
                Student t = stu[j];
                stu[j] = stu[j + 1];
                stu[j + 1] = t;
            }

    // 排名（处理并列）
    stu[0].rank = 1;
    for (int i = 1; i < n; i++)
    {
        if (stu[i].total == stu[i - 1].total)
            stu[i].rank = stu[i - 1].rank;
        else
            stu[i].rank = i + 1;
    }

    // 输出
    printf("\n====== 成绩排名表 ======\n");
    printf("%-4s %-8s %-10s %-6s %-6s %-6s %-7s %-6s %-4s\n",
           "名次", "学号", "姓名", "课1", "课2", "课3", "总分", "平均", "排名");
    for (int i = 0; i < n; i++)
```

```
        printf("%-4d %-8d %-10s %-6.1f %-6.1f %-6.1f %-7.1f %-6.1f %-4d\n",
               i + 1, stu[i].id, stu[i].name,
               stu[i].score[0], stu[i].score[1], stu[i].score[2],
               stu[i].total, stu[i].average, stu[i].rank);
    return 0;
}
```